

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Content was collected from 24 Qatari high schools with 4,372 students under exam-style supervision [Q3, Q27], with prompts deliberately designed to be 'developmentally appropriate for the target age groups, culturally relevant to the Arabic-speaking context, and free of sensitive topics' [Q99] and gender balance pursued [Q23, Q100]. However, the deployment targets Egypt, Saudi Arabia, Jordan, Lebanon and other Arab countries [elicitation A1], and the authors themselves acknowledge that single-country sampling 'may limit its generalizability to other Arabic-speaking populations due to diverse educational systems' [Q88]. The Qatar expatriate mitigation is partial only [Q88]. Web research confirms that secondary-curriculum Arabic writing rubrics in Egypt, Saudi Arabia, Jordan, and Lebanon are unverified against CAST [WEB-5, WEB-6, WEB-7], that Lebanese students show documented MSA-register avoidance [WEB-9], and that within Qatar itself ~60% of secondary students attend private schools across 23 curricula [WEB-1, WEB-3, WEB-4] with no documented school-type breakdown of the LAILA sample. Two prompts (P3, P4) are also under-represented [Q90], and average length is only 171 words [Q92], restricting extended-writing assessment. Given IC is HIGH priority, multiple substantive content gaps weigh heavily.

ID	Evidence
Q3	"Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings." (p.1)
Q27	"Data collection spanned one academic year through 27 school visits (24 initial, three follow-ups), by nine team members, to meet the target student count." (p.5)
Q99	"Essay prompts were carefully designed to be developmentally appropriate for the target age groups, culturally relevant to the Arabic-speaking context, and free of sensitive topics or potentially harmful content." (p.10)
Q23	"[D3] Achieving Gender Balance: We intentionally aimed to balance the number of essays from male and female students to promote fairness, reduce gender-based bias, support inclusive evaluation, and enhance model generalization." (p.3)
Q100	"The dataset includes balanced gender representation and covers grades 10-12." (p.10)
Q88	"First, LAILA was collected from students in a single country, Qatar, and specific grade levels, grades 10 to 12, which may limit its generalizability to other Arabic-speaking populations due to diverse educational systems. However, this concern is partially mitigated by Qatar's large population of Arab expatriates, resulting in student participants representing a wide variety of Arabic dialects and backgrounds." (p.9)
Q90	"Third, two of the prompts (Prompts 3 and 4) have fewer essays than the rest, which could introduce imbalance and lead to skewed performance in prompt-specific evaluations." (p.10)
Q92	"Fifth, with an average essay length of only 171 words, the dataset lacks extended writing samples, restricting the assessment of models on long-form academic tasks requiring more complex, extended compositions." (p.10)
WEB-5	Egypt's curriculum reform suggests evolving writing-assessment norms not aligned with CAST (https://www.impact-se.org/middle-eastern-curriculum-reform-a-window-into-national-values/)
WEB-6	Saudi Arabia uses centrally produced rating tools managed by supervisors (https://www.moe.gov.sa/ar/education/studies/Documents/Education%20in%20Saudi%20Arabia.pdf)
WEB-7	Lebanon's CRDP authoritative but Arabic writing rubrics not in searchable form (https://wenr.wes.org/2017/05/education-in-lebanon)
WEB-9	Lebanese student MSA-register avoidance suggests differing teacher norms (https://www.researchgate.net/publication/267714826_Foreign_Language_Education_in_Lebanon_A_Context_of_Cultural_and_Curricular_Complexities)
WEB-1	Qatar 2024-2025: 278 government schools (~137,048 students), 351 private schools (~228,488 students); ~60% of secondary students in private schools (https://grokipedia.com/page/List_of_schools_in_Qatar)
WEB-3	https://www.tutorchase.com/blog/the-education-system-in-qatar-explained

WEB-4	23 distinct curricula operate in Qatar; British curriculum leads at 45.8% of K-12 schools (https://consulting-haus.com/wp-content/uploads/2023/09/CH_Education_Sector-Overview_September-2023.pdf)
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Strengths

- Authentic classroom-produced writing under exam-style supervised conditions with IRB approval [Q21, Q3, Q95]
- Deliberate cultural relevance and age-appropriateness in prompt design [Q99]
- Explicit gender balance principle [Q23, Q100]
- Largest publicly available Arabic AES dataset [Q1, Q79]

ID	Evidence
Q21	"[D1] Ensuring Data Integrity: We designed LAILA to capture authentic, classroom-produced writing created without external assistance. This design choice preserves genuine linguistic variation, learner errors, and authentic complexities essential for real-world AES applications." (p.3)
Q3	"Under Institutional Review Board (IRB) approval, we visited 24 high schools in Qatar and collected essays directly from 4,372 students in authentic classroom settings." (p.1)
Q95	"The collection of LAILA received approval from an Institutional Review Board (IRB Number: QU-IRB 159/2024-EA) and was conducted with informed consent from both students and their legal guardians." (p.10)
Q99	"Essay prompts were carefully designed to be developmentally appropriate for the target age groups, culturally relevant to the Arabic-speaking context, and free of sensitive topics or potentially harmful content." (p.10)
Q23	"[D3] Achieving Gender Balance: We intentionally aimed to balance the number of essays from male and female students to promote fairness, reduce gender-based bias, support inclusive evaluation, and enhance model generalization." (p.3)
Q100	"The dataset includes balanced gender representation and covers grades 10-12." (p.10)
Q1	"LAILA, the largest publicly available Arabic AES dataset to date, comprising 7,859 essays annotated with holistic and trait-specific scores on seven dimensions: relevance, organization, vocabulary, style, development, mechanics, and grammar." (p.1)
Q79	"To bridge this gap, this paper introduced LAILA, the first large-scale publicly available dataset for Arabic AES." (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — Arabic high-school essays inherently require cultural, dialectal, and curricular knowledge. LAILA's content is anchored in Qatari institutional norms (QUTC, Qatar MoE) [Q33, Q84], and the regional YAML notes that Levantine-, Egyptian-, and French-contact-influenced MSA may diverge from Qatar's Gulf MSA conventions [WEB-9, gap_id 4].

IC-2: Partially — Qatar-targeted prompts are culturally appropriate for Qatar [Q99] but unvalidated for Egypt, Saudi Arabia, Jordan, or Lebanon, where curriculum bodies and rubrics differ [WEB-5, WEB-6, WEB-7] and no cross-national rubric comparison has been published (gap_id 1).

IC-3: No clear Western-specific content was identified; LLM selection used the Open Arabic LLM Leaderboard [Q178], and prompts are Arabic. The risk is Qatar-centric rather than Western-centric content.

IC-4: INSUFFICIENT DOCUMENTATION — the LAILA paper does not document recruitment of non-Qatari Arab annotators; the regional YAML explicitly flags this as a gap requiring expert elicitation in each target country (gap_id 1).

IC-5: Documented content issues: single-country sourcing [Q88], unverified cross-national applicability, prompt-count imbalance for P3/P4 [Q90], short average length restricting long-form assessment [Q92], and undocumented Qatari school-type representation [WEB-1, WEB-4].

ID	Evidence
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q84	"We heartily thank our dedicated annotators for their contributions and express our gratitude to Qatar University Testing Center, the Ministry of Education and Higher Education in Qatar (the Arabic Section of the Department Of Educational Supervision in particular), participating schools, and students for making this work possible." (p.9)
WEB-9	Lebanese student MSA-register avoidance suggests differing teacher norms (https://www.researchgate.net/publication/267714826_Foreign_Language_Education_in_Lebanon_A_Context_of_Cultural_and_Curricular_Complexities)
Q99	"Essay prompts were carefully designed to be developmentally appropriate for the target age groups, culturally relevant to the Arabic-speaking context, and free of sensitive topics or potentially harmful content." (p.10)
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WEB-5	Egypt's curriculum reform suggests evolving writing-assessment norms not aligned with CAST (https://www.impact-se.org/middle-eastern-curriculum-reform-a-window-into-national-values/)
WEB-6	Saudi Arabia uses centrally produced rating tools managed by supervisors (https://www.moe.gov.sa/ar/education/studies/Documents/Education%20in%20Saudi%20Arabia.pdf)
WEB-7	Lebanon's CRDP authoritative but Arabic writing rubrics not in searchable form (https://wenr.wes.org/2017/05/education-in-lebanon)
Q178	"The selection of the LLMs was based on the Open Arabic LLM Leaderboard, where we selected" (p.17)
Q90	"Third, two of the prompts (Prompts 3 and 4) have fewer essays than the rest, which could introduce imbalance and lead to skewed performance in prompt-specific evaluations." (p.10)
Q92	"Fifth, with an average essay length of only 171 words, the dataset lacks extended writing samples, restricting the assessment of models on long-form academic tasks requiring more complex, extended compositions." (p.10)
WEB-1	Qatar 2024–2025: 278 government schools (~137,048 students), 351 private schools (~228,488 students); ~60% of secondary students in private schools (https://grokipedia.com/page/List_of_schools_in_Qatar)
WEB-4	23 distinct curricula operate in Qatar; British curriculum leads at 45.8% of K-12 schools (https://consulting-haus.com/wp-content/uploads/2023/09/CH_Education_Sector-Overview_September-2023.pdf)

Information Gaps

- School-type breakdown of LAILA's 24-school sample (government vs. Arabic private vs. international vs. community)
- Whether and how non-Qatari Arab dialectal influence on MSA is treated by annotators
- Cross-national curriculum rubric divergence for STY/DEV

Requires Expert Verification

- Curriculum specialists in Egypt, Saudi Arabia, Jordan, Lebanon to confirm whether Qatari prompts and topic conventions transfer
- LAILA authors regarding within-Qatar school-type sampling

Remediation

Gap 1: All essays are from Qatari students; cross-national generalizability is acknowledged as limited and within-Qatar school-type breakdown is undocumented.

Recommendation: Conduct cross-national essay collection in Egypt, Saudi Arabia, Jordan, and Lebanon at G10–12, and audit the Qatar sample's school-type distribution against MoEHE statistics (government, Arabic private, international, community).